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Vinod Khosla and Sun Microsystems (A)

The president of Computervision was on the line. "We like your workstation," he said, "but the deal with Apollo really is done. I don't see how you could change our minds."

It was July 1983. The future of Sun Microsystems, Vinod Khosla thought, lay in the balance. He had to stop Computervision from signing a contract with Apollo. But how could he? Sun was a 40-employee enterprise, started just over a year ago by a team of 26- and 27-year-olds, that had just posted its first million-dollar sales month. Apollo, on the other hand, owned the engineering workstation business. Founded by industry veterans in 1980, Apollo provided high-performance workstations to Computervision's key competitors. Could Computervision bet its future on Sun?

The Workstation Market

Workstations, like personal computers (PCs), were designed to provide users with dedicated computing power. Historically, many users had shared the computing power of a single minicomputer or mainframe computer through more or less "dumb" terminals. Workstations and PCs, on the other hand, gave individuals their own CPUs (Central Processing Units—the "brains of a computer,") at their own desks.

There were, however, two important differences between workstations and PCs. First, workstations were designed to provide more computing power (close to a minicomputer's) and a greater variety of functions than were PCs. Bitmapped screens and graphics displays, for example, were considered standard in a workstation, but not in a personal computer. Correspondingly, a workstation sold for about $20,000, compared to a $5,000 to $7,000 unit cost for a PC. Second, while a PC system was entirely self-contained, workstations were usually attached to a network in order to share devices like printers and file servers.

The concept of a workstation—a high-performance desktop computer linked to a network—was developed at Xerox PARC[1] in California. Xerox defined the hardware specifications and the network's operating systems. It also designed the network, which it called Ethernet. Xerox first had a functioning workstation in 1976. By 1982, it was running sophisticated applications, like desktop

[1] Palo Alto Research Center.

Professor Amar Bhide prepared this case as the basis for class discussion rather than to illustrate either effective or ineffective handling of an administrative situation.

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publishing, on workstation networks. Xerox PARC was, however, ahead of its time. Its workstations were expensive and could not compete against traditional, centralized computer systems.[2]

By 1982, though, declining microprocessor and memory costs had turned the economics around. According to Khosla:

> You could buy a microprocessor with half a MIP of computing power for $100 and memory for a few hundred dollars. Yet most existing computing environments caused 30 to 50 engineers to share that amount of processing and memory. Here you were paying $60,000 to $70,000 total burden cost for an engineer and you were having them share their principal tool, memory and processors, 30 to 50 ways. The reason was because the memory and processors were attached to disks and tapes and printers that cost a few hundred thousand dollars.
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> But now, this technology called Ethernet networks coming out of Xerox PARC in Stanford gave you the ability to separate those two resources. If you look inside a computer, it has a bus; on it is the processor, memory, and the disks and tapes and all that. Ethernet sort of stretches the bus over hundreds of meters so that the cheap stuff can be dedicated and you only need to share what's expensive. So fundamentally, the rapid drop in the cost of memory and microprocessors and the availability of Ethernet changed the economic model.

Not surprisingly, workstations and PCs served different customer bases. Typical PC users were light users, in terms of time spent and performance demanded; computing was peripheral to their jobs, and before the advent of PCs, most had not been active computer users. In contrast, the typical workstation customer—often a scientist or engineer—was a “power user” who relied heavily on computers.

Computer science departments of universities constituted an important customer segment. Student enrollment in computer science boomed, but departmental mainframes and minis were stretched to their limits. Rather than adding new centralized CPUs, many universities installed networked workstations.

In industry, workstations were used in computer-aided design (CAD) and computer-aided engineering (CAE). CAD and CAE took off in the seventies, when applying computers to the design of mechanical parts, buildings, and even electronic circuits proved a major success. Initially, CAD/CAE software was tied to minicomputer hardware, but now workstation-based systems were gaining popularity, especially at the low end.

CAD and CAE users like GM and Boeing either bought hardware and CAD and CAE software separately or bought turnkey systems from the so-called OEMs (original equipment manufacturers).[3] OEMs, in turn, exhibited varying levels of vertical integration. Computervision, Daisy, and Intergraph, who were market leaders in their respective segments, produced their own hardware and software. Almost all the other OEMs, however, were "system integrators": they concentrated on developing CAD/CAE software and bought the hardware they needed from one or more vendors. Apollo's workstations were adopted by the three major OEMs that were not vertically integrated—Mentor Graphics, Calma, and Autotrol—as well as by most of the smaller OEMs.

Apollo faced little competition. Digital, IBM, Prime, and other vendors of minicomputers and mainframes—the traditional hardware “platforms” for CAD and CAE applications—did not offer

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[2] Sometimes referred to as "timesharing" architectures.

[3] Since many OEMs did not build their own hardware at all, the term is a confusing misnomer. It has, however, been firmly established in the industry jargon.

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workstations. According to Apollo's marketing director, mainframe and minicomputer manufacturers were too wedded to their existing lines of business: "Anything that they may want to incorporate has to be made compatible with their existing timesharing system, and this takes the edge off their price and performance." The only established player that had announced its intention to compete was Hewlett-Packard; in the meantime, the only firms that Apollo saw in the marketplace in 1983 were scores of small start-ups, the largest of which was Sun. These start-ups were thinly capitalized and offered workstations whose broad design specifications were all drawn from the work at Xerox PARC; they were therefore very similar.

Background

Before Sun, Khosla had been a founder of Daisy Systems, an integrated vendor of CAE systems. Khosla recalled how he got involved with Daisy:

> Since I was 15, I had wanted to start a company in Silicon Valley. When I finished my bachelor's in electrical engineering at the Indian Institute of Technology, I had tried to start a company in New Delhi to produce synthetic milk out of soy extract. This ran into utter frustration. I was 20 and very impatient. So I decided to come here and find my way to Silicon Valley. I first went to Carnegie-Mellon and got my master's in biomedical engineering. The reason was purely economic—they agreed to fund me.

Khosla then joined the Stanford MBA program. He worked for a small electronics company in the summer after his first year and, as he prepared to graduate, sought a position in a small company.

> I established a set of criteria. I would only work with companies started after 1976, and this was in 1979. I would only work with companies that had less than a hundred employees. So I went through the American Electronics Association directory and wrote a letter to every company that met these criteria. There were 400 letters I wrote, which was a big burden for me. Four hundred letters was about $400.
>
> It didn't pay off. But there is a small business club at Stanford, and I had a little team that was helping high-tech companies write business plans. One of the guys on my team said his dad was talking to some guys who wanted to start a company out of Intel. I called them, met them for about an hour, and accepted the job. This was early in 1980.

Daisy was a pioneer in the computer-aided engineering business. Its goal was to provide engineers with the means to do circuit design on computers. Khosla began to question its hardware strategy.

> We invested 80% of our resources in the first 18 months building the computer hardware needed for CAE and about 20% developing the CAE tools (the software). I thought that was ridiculous. If you are building a general-purpose computer, you might make a $10 million investment in semiconductor technology to build gate arrays or something like that, because you know that cost will be amortized. But if you are building hardware for one application and you know your volume is going to be 500 units a year, not 50,000 or 100,000 units a year, your economics are very different. You end up with a very much more expensive machine and with much more limited capability.
>
> Yet nobody was building a computer that Daisy could use, because the CAE market was so new. There were fundamentally no CAE tools available. There was a chicken and egg problem.

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Apollo had just introduced a workstation, but it didn't fit Daisy's needs. So the company developed its own hardware from scratch.

> We bought a microprocessor from Intel and built a whole machine around it that was appropriate for our CAD applications. It was clear that it had a lot of limitations. The day after we delivered our first machine, and I personally went and delivered it, the customer said, "Can I have a Pascal compiler for it?" A day after they got this complete tool set, which they paid a lot of money for, they wanted to start changing it and adding programs to it. That basically said to me that there really was a market for a general-purpose machine.

Khosla investigated whether Daisy could modify and market the hardware it had developed, but

> It never seemed to make sense, because there were too many other things to do in their primary business. This was a different business requiring a different set of criteria, a different set of support and business infrastructure.

Therefore, Khosla decided to leave Daisy in December 1981 to pursue this new opportunity.

From Concept to Prototype

Finding a Collaborator

Khosla drew up a specification for a workstation that was influenced strongly by the work at Xerox PARC, The workstation would use a Motorola 68000 microprocessor, have a bitmapped display, run the UNIX operating system, and plug into the Ethernet network. These were, according to Khosla, "the only reasonable specs," as evidenced by the fact that by 1983 "there were maybe 30 companies implementing exactly that same spec." As *Datamation* noted in March 1983: "The new venture uniform of UNIX and the Motorola 68000 is getting as fashionable as IBM's blue and white stripes."

Notwithstanding his engineering training, Khosla looked for a collaborator:

> I'm probably more of a conceptual engineer, and I can draw block diagrams for almost anything I can think of, but I can almost never implement them. So I started looking for someone who had done this kind of stuff before. I heard of a project at Stanford called the Stanford University Network, or Sun workstation project. I called the computer science department, and some secretary who did not want to bother a professor gave me the name of a graduate student from Germany, Andy Bechtolsheim.
>
> Apparently, Andy, who was also at Carnegie at the same time I was, but I did not know him there, had come to Stanford to do his Ph.D. in CAD tools. I think he realized there was no appropriate machine to develop CAD tools, following the same discovery process I had gone through, so he decided to build one himself. His specs fit mine almost to a T.
>
> Andy had developed the workstation concept in a fair amount of detail and had a prototype implementation of it. Stanford had assigned the technology to him because, in their great wisdom, and after calling DEC and Prime, they had decided it had no value.
>
> So, for over a year he had been licensing the technology to six or seven companies. He had invested $25,000 of his own money into building prototypes, and as a grad student licensing it at $10,000 a pop, he thought that was just wonderful.

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Bechtolsheim offered Khosla his usual $10,000 license. Instead, Khosla tried to persuade Bechtolsheim to join forces to start a company to build workstations based on his designs.

> I said to him, "I want the goose that laid the golden egg, and I don't want the golden egg." I thought that kind of resource is very rare to get. So I would rather have him than any one design he would come up with. I had nothing very concrete to offer. I told him we could build a big company, that we could raise a few million dollars. He would be a founder of the company.
>
> It was not a tough sell.
>
> Andy is just a very easy person to work with. He is terribly nice, very bright, very, very accommodating. His only concern was, should he drop his Ph.D. at this point? He claimed to be about three or four months away from finishing his Ph. D. I thought he was nine months away. I finally convinced him he could come back and finish his Ph.D. six months later. If he thought, in fact, that it was that short a distance, he could do it in parallel.
>
> That was the tough part of the sell, and that was very tough I think. But the rest of it was not. He had had enough frustration with these other guys who he had licensed to, not understanding his concepts even though they had all the details. These people had seen some sophisticated technology, grabbed it, but did not know what to do with it. It was almost like they had a design for a Lamborghini engine and they put it in a Volkswagen chassis, so you couldn't corner more than 50 miles an hour. They had this incredible performance design, but anything they added to it to complete it limited it rather than enhanced it.
>
> You really have to conceptually understand something. It’s got to be in your gut. Intuition. Otherwise almost all technology licenses fail to work. In fact, if you look at the history of technology licensing, by and large, it has failed. People never, never believe how important it is to have that understanding. They just fundamentally don't believe it!

Obtaining Seed Money

Andy Bechtolsheim agreed to participate by late January 1982. The two started working out of Andy's office at Stanford and in a couple of weeks had produced a brief plan (see **Exhibit 3**).

> It was a real concise statement of the reasons for making an investment: how the economics had changed, what the product would be, when it would be out, and how big it could be and why the market made sense.
>
> The next day, February 12, we met with two venture capitalists, one of whom, Bob Sackman, had helped me write the Daisy business plan. Within three or four days, they agreed to give us $300,000 in equity. They gave us a $100,000 check right away and said, "You can get going and let's work on the paperwork." On February 22, we formally incorporated the company and received the remaining $200,000. The price of the stock was $2.75 a share. We also gave them an option to put an additional $2.2 million for a total of $2.5 million at $5.60 a share, that option to expire on June 30, 1982. By that date we were supposed to hire a marketing person, write a business plan, and demonstrate a prototype.
>
> Bob Sackman led the thing, and he trusted me. It was really on trust. There was very little due diligence on their part—they just believed in the concept and said, "Yes, we think you can do it."

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Getting Started

Sun failed to recruit a marketing manager by June 30. Khosla explained:

> I made organization charts for every major company I could think of, and I made over a hundred phone calls. Most of them just hung up on me. When you are 26 years old, look like a little kid, talk with a funny accent, and have two people in your company, and you're calling the vice president at DEC or HP, you don't get very far. . . . Besides, you can't hire someone until you have the money to pay them. Nobody wants to leave established companies to join someplace where they can't even see six months of salary.

Progress on other fronts, however, was rapid. Within days of obtaining seed financing, Khosla had persuaded Scott McNealy, a friend from his MBA class, to leave his job as director of operations at a small hightech firm called Onyx and run manufacturing for Sun. By May 20, Sun not only had produced a prototype, it had also made its first sale.

> We had to procure power supplies, procure monitors, select our vendors, design the housing, and design sheet metal. Andy and I designed the sheet metal. So we became industrial designers for a while and I became the purchasing agent to buy all these components, do component selection and all that. Then there was a bunch of software to be developed, because these things are worthless without software, so what we did was develop some terminal emulation software. You could at least do a terminal emulation and appear to a computer as a terminal with some graphics capabilities. And we promised our customers that we would soon have UNIX [an operating system] running on it.
>
> Our first customer was another start-up called Solo Systems. They had requirements for a box just like ours for reasons similar to why Daisy needed such a box. They didn't really need much software from us since they were going to develop their own operating system and applications software.

Recruiting Bill Joy

Unlike Solo, other customers demanded that Sun provide an operating system with its hardware. The original specifications had called for the UNIX operating system; Sun now needed to decide which kind of UNIX to provide. As a stopgap the company contracted with a software house, Unisoft, to produce a quick version of UNIX. For the longer term, Sun appeared to have two choices: AT&T UNIX or Berkeley UNIX.

> It was the biggest decision I had to make last year. AT&T had developed UNIX, but they didn't take it seriously; nobody did. Their version matched the old 16-bit minicomputer hardware but not the new 32-bit machines. The Berkeley UNIX, on the other hand, had been optimized for 32-bit hardware. Berkeley's UNIX also came with a wide range of utilities, contributed by a variety of universities, that enhanced its power and usefulness. But while Berkeley UNIX was technically superior, AT&T was the industry standard, to the extent that there was a standard.
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> There was no question in Andy's mind that Berkeley UNIX was the right thing to do, but in my mind, I had to play out the business issues. Everybody else was going with AT&T. Everybody.

Khosla picked the Berkeley version.

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As he did with Andy Bechtolsheim, Khosla tried to recruit the resident UNIX guru at Berkeley, Bill Joy, rather than merely license the software. At age 26, Bill Joy was a UNIX star.

> When he called DEC and said "I need a VAX," they used to say they'd deliver a six-pack of VAXes, free. DEC knew that Bill was responsible for a lot of their university sales. Berkeley UNIX was getting to be very popular in the universities, DARPA,[4] and the defense community, so Digital really wanted this work at Berkeley to continue, and so I think he had 10 to 12 VAXes in that one room all to himself.

By June, Bill Joy had been persuaded to join Sun. His reasons, speculates Khosla, were similar to Bechtolsheim's.

> I think he joined mostly to make the Berkeley system a commercial product. Andy didn't have a lot of software interest or expertise or didn't want to do anything about it. When Bill Joy saw the hardware, he saw a way to influence the hardware to match his operating system and to bring the two together. The financial issues were not small, but I think his ego played a big role.

The participation of Bechtolsheim and Joy enabled Sun in the months ahead to build top-notch engineering and software teams.

> Most of the engineers came because they wanted to work with Bill Joy and Andy, who were leaders in the field. They also wanted to work on commercial products that could be leadership products. They were all excited about this kind of stuff because it was well publicized in the research community by Xerox PARC.
>
> These "worker bees" weren't as conservative as the senior managers who wouldn't talk to us. In fact, the worker bees considered management in their existing companies a problem. It was the same managers that didn't want to come to work for us.

The Business Plan

Key Strategic Principles

In parallel with his other tasks Khosla began writing the business plan. In this endeavor, he had little help.

> Anybody you really tried to talk to outside started with "You are foolish." Inside, Scott, who had never used a computer before, knew nothing of the technology or the market or the strategies. Andy and Bill didn't either, except they understood what things could be used for very, very well. Bill and Andy really had a very great perspective on what was usable and what wasn't. Bill and Andy are the best marketing people in terms of specifying the marketing requirements, because they know what technology can do and know what people need to do. They are dealing with technical scientific people whose needs are similar to their own. That was a big help, because I knew it globally but not quite as intuitively and in as much depth as these two guys did. But they had no interest in the business end of things. The business selection of technologies was my responsibility. As I sat down to write the business plan, I was sort of the marketer, the marketing strategist, sort of the technology person.

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[4] Defense Advanced Research Projects Agency.

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By the end of June Khosla had completed the business plan. While much of it described the operational plans, it also laid out Sun's key strategic principles.

> We said, "We want to take on IBM and DEC." And not only do we want to take on IBM and DEC, we want to take them in the mainstream business, not in the little niches that most companies had positioned themselves in the Valley.
>
> We had big dreams. When I was hang gliding, I remember seeing this motto: "Success comes to those that dare to dream dreams and are foolish enough to try and make them come true." You have to try doing something extraordinary, because unless you try something extraordinary, you won't ever do anything extraordinary.
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> We were going after a general business. We wouldn't develop any applications software. We would not specialize in any one narrow function—whenever we had a trade-off we'd go after the general computing environment. For example, should we use our resources to build better compilers in the UNIX development environment, or should we build a fast graphics thing like Silicon Graphics was doing? We'd pick the general one, which we'd need to compete against DEC, as opposed to the one we'd need to compete against Megatek or Lexidata. We chose DEC as the target even though we knew it was a much harder target to get than Lexidata or Megatek.

Sun's ambitions led to the choice of a direct sales force.

> There was discussion on distributors or direct: how do you sell and support it, how about advertising? All the little things which we had no money for. I made the decision that I only wanted to do a direct sales force, because that was what all the successful computer companies had. I did not know anybody who had been successful through third-party distribution and third-party support.
>
> It was very hard, because you didn't know how you were going to get revenue with a direct sales force, and you knew on the distribution side you could probably pick up revenue fairly quickly and not have to make investments with money you did not have in the first place. But "the moon or bust" was our motto. As Scott said, "If we are going to flop, we might as well be controversial and be a big belly flop."
>
> Our calculations showed we would not have money for anything besides a direct sales force and direct support. So marketing went out of the window at Sun. We couldn't afford any promotion, we couldn't afford a PR firm and all those things, so we did without those. If we did not get money to put a color brochure together, that was OK. Because we weren't going to compromise the key fundamentals that were going to make us successful.

The business plan reaffirmed Sun's commitment to standard components. Andy Bechtolsheim's prototype and initial design had employed off-the-shelf building blocks because he had no choice.

> Stanford had paid for some very early development, but once Andy got past the bread-boarding stage they did not want to pay for it and he funded it out of his own pocket. So he picked what was available: Ethernet cards from 3-Com, memory and disk controller cards for the multibus, and the like.

Khosla believed that Sun should continue this strategy.

> I looked at what Sun needed to be competitive and I knew I wouldn't have the resources. I picked the smallest successful entity I could find which was Hewlett-Packard, which was

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developing some entries in the technical side of the computing arena. I said I need to have development resources comparable to this operation, because that has to be critical mass. Since they are stable and everybody below them I don't consider stable, my fundamental objective is going to be to get as close to them in resources as quickly as possible.

They had 200 engineers and were going to spend $300 million on their workstations. I figured, if I got 80 or so engineers I could do the same work as 200 HP engineers could. But I could find no way to raise even those 80 engineers.

The only way was to convince other people to do your development for you. The only way to get people to do development for this hole-in-the-wall, five-person operation would be if they were doing the development anyway for somebody else. And the whole idea of standards came because there were other people doing other things: multibus disk controllers, multibus serial I/0 cards, standard memories, standard Ethernet chips. Ten people here, 5 people here, and I could add up to 80 people that I could get relatively quickly working on my project.

It was absolutely counter to any business principle in any computer company. We were severely criticized for being so standard. Everybody said that with this standard product with a bunch of things thrown together, you will never have a proprietary advantage.

Apollo, the industry leader, had taken the opposite tack. Whereas Sun had adopted Xerox's Ethernet, Apollo had developed its own Domain network. Whereas Sun had adopted Berkeley UNIX, Apollo had invested in its own operating system.

They took the conventional view and chose not to have standards. In fact, I talked to their marketing guy who I was trying to recruit about using standards and he said, "What a stupid idea." He said, "I have just finished going through my literature and eliminating all references to the Motorola 68000 chip because people might perceive it as a standard."

Manufacturing Strategy

Sun would depend on outside resources for manufacturing.

We would do as little as possible because of resource constraints. And our costs would be lower because whoever was doing our board manufacturing would amortize 2% or 3% of the overhead of managing such a facility to us, while if you were doing it yourself you amortized 100%. Even after you added in a 10% or 15% profit margin we were still better off than doing it on our own.

The company would rely on the network of suppliers in Silicon Valley.

There is a lot of infrastructure here. You can get PC boards turned around in a week. There are a lot of companies here that only support small companies. You can get almost anything you want. You get more than you can get internally in a large company: In terms of resources, facilities, turnaround on components, pricing on components, response time on bids, range of services—from mechanical design to custom fabric machining to PC board fabrication to component loading. If there's an expensive piece of equipment like a $2 million tester, somebody probably has it and is amortizing it across 20 companies. And all these guys are used to dealing with small companies, used to being responsive, because their business depends on it.

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In contrast, Apollo, which was located on the East Coast and designed its own proprietary components, was much more vertically integrated in manufacturing.

Target Customers

Sun's long-term goals for obtaining customers had to be tempered by short-term realities.

> The issue was, who would buy these things, half-baked as they were, and yet be important customers? We did not want just revenue. We wanted strategic revenue.
>
> There were two classes of people who could use a not-fully-developed machine. Universities could use it, because they did all kinds of kludgy stuff themselves and they did not mind putting in resources. Typically, they were trying something new so they did not have to rely on the whole hardware as much. Second, OEMs could use it. Daisy, Computervision, and others could essentially develop a machine from scratch, and they could definitely develop it from halfway on.
>
> It was clear that not many OEMs would buy it. We had no credibility. So we were only left with the universities as a realistic customer set for our first year.
>
> Prior to us, DEC had been successful in establishing itself primarily through the technical credibility they gained with universities. Whenever a commercial customer went to a university for consulting help, they talked about DEC, because those were the state-of-the-art machines. So in the first year I decided to set a sales target of the top 50 computer science departments in the United States. The next year we would go after OEMs.

Venture Capital

As June 30 approached, Sun's founders decided to broaden their base of funding. They persuaded the two firms that had provided the seed capital not to exercise their option to put $2.3 million of stock while Sun sought to raise funds from three highly regarded venture capitalist firms--Kleiner Perkins, TVI, and TA Associates.

> The original venture capitalists were very cooperative in not exercising their option. They said if the other firms, which were real brand name firms, were involved it would add a lot to the credibility of the company.
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> TA, a Boston firm, refused to participate.
>
> They had just opened their office, and the guy who started the office had been in the same MBA class with Scott and me. And there was this real paranoia, a concern with perceptions.

The other two firms were attracted to the concept but had reservations about the team.

> They thought the business plan was very good. Experts in universities they checked with were enthusiastic about the future of workstations. And we had found customers for our early units. We had generated revenue.
>
> On the other hand, none of us really had any meaningful work experience. Andy and Bill had been grad students. Scott had worked at FMC for a year and had managed a manufacturing operation for a small high-tech firm thereafter. I had never really worked at a

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serious job. I had been in one start-up for a year and a half, but that was it. I understood the technologies but I didn't understand a lot of things about doing business.

One of the venture capitalists at TVI insisted that Sun add more experienced members to the team. The founders, whose earlier efforts at recruiting had not been successful, balked. The disagreement caused the financing to be delayed until September, although some funds were made available earlier. Sun raised $4.5 million—$2.3 million from the seed investors and $2.2 million from Kleiner Perkins and TVI. The seed investors got slightly cheaper stock.

The First Year: June 1982 to June 1983

The University Market

In the first year of full operations, Sun succeeded in developing a strong presence in the university market.

> We all worked hard to sell to universities. Our rule was, if anyone was traveling, they had to call on a university. That was very, very fortunate, because universities were really looking for a machine like this. They could really play with it because it was standard, and they could buy standard pieces of hardware and interfaces, and they liked that a lot. And nobody else had anything close. The universities weren't interested in Apollo, and Apollo wasn't interested in the universities. It was small peanuts revenue.
>
> Stanford, Berkeley, and Carnegie-Mellon were our early breaks. They were three of the top four computer science schools in the country, and that counted for a lot. All the other computer science departments—42 out of our target 50—then picked it up around the country. So in our first fiscal year starting July 1, 1982, we have been able to do $8 million of business, 80% of it at universities.
>
> Universities have become our mechanism for credibility. It had worked for DEC, and it is working like a charm for us. Whenever Berkeley or Stanford talks about computing in the future, they talk about Sun, and it is the gospel message coming from an impartial source. Suddenly, every OEM is calling on us because they are hearing from the top research departments around the country that we are the next generation. Although we do not have business credibility, we have technological credibility.

Sales and Support

Success in the university market allowed Sun to recruit a small sales force of three to five salespeople. Recruiting them was easy.

> They don't have career loyalties long career plans. Several came out of XYZ Computers.[5] XYZ doesn't have a corporate philosophy of long-term relationships with customers. It is, "get as much from the customer, screw him and move on." So their salespeople were really mobile.
>
> And they thought they could make a buck at Sun. The product was hot, and, unlike Digital's or IBM's, our salespeople were highly commissioned. If they did well, they could

[5] Disguised name of a minicomputer company.

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> make $200,000 to $300,000 in commissions. Anyplace where there are a few stories of people making in the hundreds of thousands of dollars selling, people just run to it. The next year they can move to another company.

Since the salespeople were "process people and not information contact people," they worked in tandem with presales support engineers, who answered most of the questions. Sun's support engineers were UNIX specialists, hired out of universities and from "forgotten groups" who had been hacking away at UNIX at large corporations like Amdahl, National Semiconductor, and Intel. "We had a pretty competitive UNIX support function from the beginning," Khosla said. "Since UNIX wasn't popular, nobody else had anything."

Khosla considered Sun's sales support engineers to be a critical resource.

> This is very much a salesman support-oriented company. The system, and especially our UNIX, is too sophisticated to hand out to somebody and walk away. Every field office has technical support people who are really UNIX programmers. That's helped a lot.

Engineering: Hardware and Software

The hardware engineering team led by Andy Bechtolsheim had its hands full in Sun's first year. Not only did it have to work on completing features Sun had promised its customers would be provided later, but it also began work on a next-generation product, the Sun 2, in October 1982. Sun 2 was scheduled to be shipped in fall 1983.

Bill Joy's software engineering team went to work adapting Berkeley UNIX to Sun's hardware as well as enhancing its capabilities.

> Initially, anybody could have been our equal, but now we are way ahead. We have 50 man-years of software invested in it. We've added a network file system and all kinds of other stuff. We have taken the utilities that were contributed by various universities and significantly improved them. So when someone buys a Sun they get what appears to be a standard operating system, but it is in fact pretty significantly added to.
>
> Our UNIX is making Sun attractive to software developers. They develop their software on our machine and then port it to other hardware. . . . When the UNIX trade show was held early this year [1983], one of the most important sessions was "News from AT&T, DEC, and Sun." That was the absolute vindication that, in the UNIX marketplace, we are going to be very important.

Financial Performance

Sun turned profitable in the first quarter, starting July 1982 and stayed profitable for the rest of the fiscal year. Khosla ran a tight ship.

> We didn't have a big overhead structure, because we didn't have any money to spend. We matched our expenses to our revenue real tight. You might ascribe this to my Indian heritage—$100K was a huge sum of money. I insist on daily cash flow reports. On the first anniversary they presented me with a sharp pencil, because, one, I needed it, and two, that was all they could afford, they said. But that is pretty much the culture of the company. We really have tight controls. Accounting, inventory control, auto management, MRP, general ledger,

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accounts payable, accounts receivable: All this was computerized by Scott and me almost from day one.

We were unlike most start-ups. Most start-ups have everything—marketing, sales, support, advertising, and PR—in place even before they have a product to sell. They get up to $600,000 to $800,000 a month of expenses before they've really started selling anything. In that range, given you're starting out with low gross margins because your product costs are high, you've got to start selling $1.5 million worth just to break even in a month.

Sun's profitability began to help it attract some experienced managers.

Since we've finished the first year, have gotten to a million dollars a month, and are profitable, people have started to look at us seriously. Started to consider us viable. People feel a lot more comfortable with something there and running and profitable and operating, even if it is small relative to nothing there. There's a radical nonlinear function between those two.

The OEM Market

One area in which Sun did not make much progress in its first year was the OEM market. Sun considered gaining share in the OEM market to be a major, long-term goal and adapted its product and pricing strategies accordingly.

One fundamental part of our strategy is to get as many CPUs out there as possible, and get them popularized. Market share is very important to us and revenue is not. Even though we did really good in revenue last year, our goal was not revenue.

So if somebody is better off buying our CPU and putting his own disk drives and tape drives, because they can't afford ours, we will let them. We have created sort of an airline pricing structure. People who want full support and service and don't want to hassle with it pay full price. People who are sort of willing to book two weeks in advance and sort of leave on Thursday evenings get a price break. Everybody puts in their level of pain to get their price point. Our goal is to skim as much of the money and also get as many of our CPUs and operating systems out there as we can. We can capture the rest of the system revenue later. And if our co-developers—all these other guys who put in the other 70 or 80 engineers in various companies—get a lot of revenue, they will invest more in Sun.

In spite of Sun's efforts, and even with the endorsement of computer science departments, OEMs had proven a very tough nut to crack.

Daisy is the leader in its market segment, and Computervision and Integraph are leaders in their market segments. These three have decided that they are going to build their own computers. Calma, Autorol, and Mentor Graphics are the second tier, but the most important second tier after the top three, and Apollo has them all. Calma, Autorol, and Mentor account for 60% to 70% of Apollo sales. Mentor, I think, accounts for 40% of Apollo and is really catching up on Daisy, and its growth has been very good for Apollo. Apollo has been going through the roof because of these customers. And because of that, all the smaller people in the business are picking Apollo.

Apollo is the successful platform. They have this huge momentum. They have $20 million or $30 million of financing versus our $4 million. They have well-connected senior

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management.[6] We have diddly squat. We clearly have the better technology, and people recognize it, but. . . .

Apollo's customers were, apparently, satisfied customers. As Mentor Graphics, Apollo's first OEM customer noted in its prospectus:

> The use of Apollo hardware enables the company to concentrate on applications software and special-purpose hardware development, while benefiting from Apollo's research and development efforts in general-purpose hardware and operating systems software. The company believes this strategy enables it to stay at the leading edge of technology in computer hardware, operating systems, and applications software. In addition, the company has benefited from market acceptance of the Apollo computer for general applications, as well as rapid improvements in the price and performance of Apollo's hardware.

As *Datamation* had noted: "Everything seems to be going like clockwork for young Apollo; the original business plan [which] called for launch in 1980, sales in 1981, profits in 1982, and a public offering in 1983 [is] right on schedule."[7]

The Computervision Order

In December 1982, Sun saw an opening into the OEM business. Computervision announced that it was considering buying a workstation instead of building its own.

Computervision was a company in transition. From the time it was founded in 1969, it had followed an ambitious strategy. It offered complete "turnkey systems"—all the hardware, software, and training that users needed. Instead of designing specialized systems for one or two industries, Computervision aimed for a wide range of domestic and international customers. And unlike most of its competitors, who added software to Digital's or IBM's minicomputers, Computervision designed and built its own.

In the seventies, according to a New England Business article, "Computervision was a high-tech company on a roll. Everything was in superlatives. It was *the* fastest growth, *the* newest technology, *the* highest income."[8]

Although Computervision was still the market leader, the CAD pioneer's position had begun to erode by 1983. IBM, Digital, Prime, Perkin Elmer and several other minicomputer manufacturers had entered the market with software from third-party vendors, and they were enjoying some success. IBM had become the number two player, almost overnight, with the help of Lockheed's popular software. And the effectiveness of Computervision's response had been undermined by delays in the development of its next generation (32-bit) minicomputer.

In addition, dozens of start-up companies were gnawing at low end of the market with cheap workstation-based systems aimed at specific niches. Using hardware from companies like Apollo and Sun, these start-up companies were able to sell CAD systems for under $100,000, compared to the $500,000 typical of Computervision's systems.

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[6] Apollo's chairman Poduska had been a co-founder of Prime; four of the seven founders were "repeaters" --they had previously founded another venture.

[7] *Datamation*, March 1983, p. 79.

[8] *New England Business*, April 18, 1983, p. 46.

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Computervision was taking several actions to shore up its position. In August 1982, founder Martin Allen gave up the titles of president and CEO to James Berret, formerly a group vice president at Honeywell. Allen said that he had "wanted someone who came from a multibillion dollar environment and who had experience competing against IBM.[9]

The company also seemed to be edging away from its commitment to vertical integration: it had bought Cambridge Interactive Systems, which sold CAD software to go with other vendors' hardware; it was negotiating to buy IBM's System 4300 computer line, as a backup to its own 32-bit minicomputer development; and, to respond to its new low-end rivals, it was planning to buy a workstation platform instead of developing one.

To gear up for the Computervision contract, Sun hired a veteran executive and group operations manager, Owen Brown, away from Digital.

> We needed credibility. The East Coast establishment didn't take me or Sun too seriously. They thought we were quirky . . . because of things like . . . I hated smokers, and for the first 20 people we didn't hire a smoker or anyone who drank coffee.
>
> Besides there was a real crunch on my time. I was chief salesman, CEO, and everything. Engineering was reporting to me. Finance was reporting to me. So we asked Owen to be VP of sales and marketing. He said, "Give me the title of President so I can deal with these Computervision guys and do big corporate account selling and go play golf with them." I didn't play golf, and in some situations it is very important. So we hired him and formed an Office of the President.

In the months that followed, competition for the Computervision order narrowed to Apollo and Sun. “We had our friends in Computervision,” said Khosla, "who were mostly the technologists, and Apollo had theirs, who were mostly the business people and all the senior executives. And there were some people who wanted Computervision not to go outside at all."

At 4:00 one afternoon in July, Khosla received an ominous call.

> It was from a purchasing guy from Computervision and that was terrible. You never want to hear from a purchasing guy, because that means you are getting a rejection. He thanked us very much for bidding on the contract but said they had chosen another vendor.
>
> I was livid. Had smoke coming out of my ears. Owen Brown, who was supposed to be running that account, was off for two weeks on his Navy reserve training. How could he be off on Navy reserve training or anywhere else when the future of the company depended on this one deal and he didn't even know the deal is happening in this two-week period?
>
> I took over. By 6:00 p.m. I had sent off a letter, Federal Express, to about 30 or 40 people at Computervision. I said we would do anything for their business.
>
> I didn't go home; I had my wife bring my clothes to the office, and I caught a red-eye to Boston.
>
> The next morning, I was in the Computervision lobby, making phone calls, trying to see someone. Nobody would talk to me.

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[9] *Business Week*, December 20, 1982, p. 76C.

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After four hours and over 50 phone calls, I got through to the VP of sales and marketing, who was a Sun friend. Apparently, Computervision had made the decision long ago. By now, there were Computervision technical people from Europe who had arrived here for technical training, at Apollo. It had gone that far. They were on crossing the T's and dotting the I's on the contract. So people were avoiding me for political reasons.

The sales VP said that "if you go back to the Sun office in Boston, I will make sure somebody calls you and talks to you." With that promise, since I wasn't getting anywhere, I went back to the Sun office there. I showered and shaved there and waited.

The Call

Finally, the president called.

He was saying, "'We have decided, and here is why." He was giving me all the good reasons. "You are a 40-person company, and you have an incomplete product. We love your technology, but there is no way you can supply it. Apollo is the standard in the industry, well financed and well managed."

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Exhibit 1 Selected Financial Data (in thousands, except per share amounts)

	January 2-June 30, 1983	February 24-June 30, 1982
OPERATING DATA:		
Net revenues	$8,657	$86
Cost and expenses:		
Cost of sales	4,486	56
Research and development	1,868	99
Selling, general and administrative	1,715	61
Total costs	8,069	216
Operating income (loss)	588	(130)
Interest income (expense), net	136	–
Income (loss) before taxes	724	(130)
Provision for income taxes	70	–
Net income (loss)	654	(130)
Net income (loss) per share	0.04	(0.02)
Weighted average common and common equivalent shares outstanding	$14,660	$5,619
BALANCE SHEET DATA:		
Working capital	$3,357	$(14)
Total assets	7,733	405
Short-term borrowings	111	–
Long-term obligation, less current maturities	749	–
Shareholders' equity	4,849	158

Source: IPO prospectus

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Exhibit 2 Apollo: Selected Financial Data

The following table summarizes certain selected consolidated financial data and is qualified in its entirety by a more detailed Consolidated Financial Statement included elsewhere in this Prospectus. The information as of July 2, 1983 and for the six months ended July 3, 1982 and July 2, 1983 is unaudited but, in the opinion of the management of the Company, reflects all adjustments (consisting only of normal recurring adjustments) necessary to a fair presentation. The operating results for the six months ended July 2, 1983 are not necessarily indicative of the results to be expected for the full fiscal year.

Consolidated Statements of Operations:

	February 13, 1980 (inception) to	Fiscal Year Ended		Six Months Ended	
	January 3, 1981	January 2, 1982	January 1, 1983	July 3, 1982	July 2, 1982
Net sales	–	$3,400,000	$18,099,000	$5,281,000	$31,728,000
Costs and expenses:					
Cost of sales	–	2,244,000	8,047,000	2,705,000	12,940,000
Research and development	582,000	1,674,000	3,005,000	1,250,000	3,407.000
Selling, general, and administrative	644,000	2,889,000	6,842,000	6,367,000	9,652,000
Total costs and expenses	1,226,000	6,807,000	17,894,000	6,367,000	26,035,000
Operating income (loss)	(1,226,000)	(3,407,000)	205,000	(1,086,000)	5,693,000
Interest income, net	149,000	415,000	102,000	60,000	1,222,000
Income (loss) before provision for income taxes	(1,077,000)	(2,992,000)	307,000	(1,026,000)	6,915,000
Provision for income taxes	–	–	–	–	2,490,000
Income (loss) before extraordinary tax benefit	$(1,077,000)	$(2,992,000)	$ 307,000	$(1,026,000)	4,425,000
Extraordinary tax benefit from prior year net operating loss carryforwards	–	–	–	–	1,950,000(2)
Net income (loss)	$(1,077,000)	$(2,922,000)	$ 307,000	$(1 026 000)	$ 6,375,000
Earnings (loss) per common and common equivalent share:					
Income (loss) before extraordinary tax benefit	$ (.38)	$ (.69)	$ .02	$ (24)	$ .23
Net income (loss)	$ (.38)	$ (.69)	$ .02	$ (24)	$ .32
Weighted average number of common and common equivalent shares outstanding	2,819,000	4,349,000	17,100,000	4,299,000	19,651,000

Consolidated Balance Sheet Data:

	January 3, 1981	January 2, 1982	January 1, 1983	July 2, 1983
Working capital	$3,178,000	$5,093,000	$ 8,176,000	$62,278,000
Total assets	3,743,000	8,579,000	19,450,000	80,247,000
Long-term capital lease obligations	226,000	914,000	2,092,000	1,838,000
Stakeholders' investment	3,332,000	6,427,000	11,761,000	69,574,000

Source: Apollo Prospectus

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Exhibit 3 VLSI SYSTEMS INC., Preliminary Business Plan (confidential)

Mission

Develop, manufacture, market, and support graphics workstations for OEM CAD/CAM marketplace. Evolve a family of compatible graphics workstations. Maintain lead with the best cost/performance product on the market.

Objectives Over Next Four Month: February 1982 - May 1982

1. PRODUCT. Bring product to market. Make first customer shipments of the SUN workstation by May 31, 1982.

2. DEVELOPMENT. Define a product family for the OEM market, including enhancements, options, and software for OEMs. Begin development of distributed UNIX operating system.

3. PEOPLE. Assemble a team of people to lead the company through its rapid growth. Recruit high-level marketing and software expertise.

4. PLANNING. The following pages contain a preliminary business plan designed to present the overall picture. A full operating plan and budget will be worked out by May 31, 1982.

5. FINANCING. Estimates for seed financing are outlined in Appendix B. Objective is to obtain seed money in February 1982 and full funding by May 31, 1982.

Tentative 2-year Plan

Following the launch in June 1982 we plan to achieve the following objectives:

1. SALES. Deliver 500 units in the first year at an ASP of $8,000 for total sales of $4M in Fiscal 1983. Ship 1500 units in the second year for sales of $10M.

2. OPERATIONS. Set up an operating company with 50 employees by the end of the first year and 150 employees by the end of the second year. Implement proper manufacturing and testing procedures, financial and accounting controls, and an international marketing and customer support organization.

3. FINANCING. Manage company growth to produce break-even cash flow by the end of the first year. At that time it is expected that $750,000 will be required to finance inventory, $1,000,000 to finance accounts receivables, and $500,000 to finance cumulative losses and capital equipment. The balance of the expected $2,500,000 financing together with receivables financing will provide working capital. Appendixes C and D list the development and staffing requirements over the next year.

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Exhibit 3 (continued)

Product: The SUN Workstation

SYSTEM. The SUN workstation is a powerful modular network-based graphics workstation. Its primary use is as a single-user computer with high resolution high performance graphics capabilities and substantial local processing power. Its modularity permits it to be reconfigured for a variety of applications, including support for networks of SUN workstations in the form of printer servers, file servers, terminal concentrators, and gateways. Such a system provides a suitable hardware base to support design automation, advanced text processing, office automation, distributed systems, computer aided manufacturing, robotics, and many other interactive graphical computing tasks.

Adhering to popular industry standards - the Motorola 68000, the Intel Multibus, the Xerox Ethernet, Bell Lab's UNIX operating system, the C, PASCAL and ADA programming languages, the SUN workstation makes it as simple as possible for an OEM to integrate the SUN workstation into his system.

HARDWARE. The SUN workstation consists of a bitmap display, keyboard, network connection, and processor. A "mouse" pointing device may be connected to the keyboard. The display has 1024 by 800 pixel resolution and can show arbitrary raster images, thus permitting variable width fonts, foreign alphabets, mathematical symbols, vectors, curves, shaded regions, and even photographic pictures. The processor is based on the Motorola 68000 CPU and provides full virtual memory management hardware. The design allows a 10 MHz 68000 to operate at full speed without wait states. The SUN workstation uses Ethernet as its local network. The Ethernet connection allows many SUN workstations to be tied together to exchange messages and electronic mail and to share services such as file storage and printing.

SOFTWARE. Three commercial versions of the Bell Lab UNIX operating system are available for the SUN workstation: Microsoft Xenix, Unisoft Unix, and Lucasfilm Unix. Commercially available PASCAL, FORTRAN, and ADA compilers have been demonstrated on the system.

Summary of Key Competitive Advantages

- Lowest Cost graphics workstation for scientific-engineering-CAD marketplace.
- High-resolution, high-speed graphics capability for text and lines.
- Design based on emerging industry standards: 68000, Multibus, Ethernet, UNIX.
- Lowest chip-count, therefore high reliability
- Proven hardware design, available today.

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Exhibit 3 (continued)

Market: OEM Workstations

The company intends to focus initially on the OEM market for high-performance graphics workstations with computational and networking capabilities. Target customers are all companies serving the turnkey CAD/CAM, robotics, process control, simulation and modeling, and other specialized industrial markets.

Currently most CAD/CAM workstations are priced in the $40,000 to $200,000 range. At this price it is difficult to justify one workstation per engineer or professional. The SUN workstation breaks new cost/performance barriers with a hardware price of $5000 to $7000 in OEM quantities. This will enable OEMs to sell their workstations at an end-user price of $10,000 to $15,000.

The SUN workstation has been reviewed and acclaimed by the scientific community. Literally hundreds of inquires have been received without any active marketing. Cadlinc, a company licensed to manufacture the SUN workstation for their turnkey CAD system has expressed a strong interest in entering the OEM market and projected sales of 1500 workstations at $7500 ASP in the first year.

This tremendous interest is prompting us to launch a new company whose specific mission is to exploit this market. The proposed company, lead by the original designers, is in the best position to take this product immediately to market. Concurrently, a family of hardware configurations and support software will be developed to provide the OEM with a range of options. Support software will be targeted to aid OEM software development. A cost reduction effort is planned to introduce a lower cost product by November, 1982.

Summary of Marketing Approach

- Focus on OEM sales for maximum growth.
- Target initial production at OEM customers for software development.
- Put SUN workstations into selected universities to gain visibility.
- Develop options/cost reductions as market requires.

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Exhibit 3 (continued)

Competitors

There is external and internal competition. External competitors are:

- Workstation manufacturers: Apollo, Three Rivers, and Xerox.
- Graphics System manufacturers: Megatek, Lexidata, RamTek, Aydin.
- Minicomputer manufacturers: Data General, DEC, HP.
- Personal Computer manufacturers: Apple, Fortune, Tandy.

Primary competition is the workstation manufacturers.

Apollo Computers Inc. The Apollo Domain product has similar capabilities to the SUN workstation: 68000, high-resolution graphics, networking. Compared to the SUN workstation, the Apollo has the disadvantage of a non-standard operating system (DOMAIN), non-standard network technology, and a retail price of $25,000 per workstation.

Three-Rivers Computer Corp. The PERQ workstation product again has similar capabilities to the SUN workstation. It is based on a bit-slice computer architecture and is difficult to use because of the limited software available. Unit price is in excess of $30,000, with pricing flexibility limited by high manufacturing costs.

Xerox Corp. The Xerox Star workstation has very similar capabilities to a SUN workstation, with identical networking and graphics capabilities. Xerox is targetting the Star to the office automation market but might also being selling into the OEM market. However, the Star system is based on a bit-slice, microcoded processor that is difficult to use for an OEM. Also, the only software Xerox offers is a proprietary programming language called MESA.

Internal competition are companies that have been licensed to build SUN workstations or SUN workstation components, usually for a special turnkey application. These companies are: Cadlinc, Codata, Forward, Imagen, and Pacific Microcomputer.

Cadlinc is a Chicago-based startup CAD company. The license agreement with Cadlinc was entered with their intention to use the design solely for a turnkey CAD workstation. With the publicity surrounding the SUN workstation, and Cadlinc being the only manufacturer of the workstation so far, Cadlinc is now planning to enter the OEM workstation market since they see an opportunity of selling 1500 SUN workstations in their fiscal 1983. Cadlinc has currently no marketing, sales, or manufacturing in place for this market. It is believed that Cadlinc can be legally delayed from entering the OEM market.

Codata is a $3M Sunnyvale company selling standalone UNIX systems based on the SUN 68000 board. They have no license to other parts of the system.

Forward Technology is a startup Santa Clara company currently selling the SUN 68000 board and the graphics board, but intending to move into the workstation marketplace.

Imagen builds the SUN 68000 board as their laser printer controller. They are not selling the board.

Pacific Microcomputer is a very small San Diego company selling the SUN 68000 board.

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Exhibit 3 (continued)

Patents and Other Rights

The original SUN workstation design has been performed by Andy Bechtolsheim while being a student at Stanford University. Subsequently, Stanford has released all their rights to the design to Andy's company, VLSI SYSTEMS INC.

The SUN workstation design includes sufficiently novel ideas that we have filed invention disclosures on two aspects of its design: the graphics subsystem, and the processor memory management.

No patent applications have been filed yet. Patents need to be filed before April 30, 1982, since the concepts were first published one year ago that day (publication bar).

The design itself has been copyrighted, and the engineering documentation is maintained as a trade secret by VLSI SYSTEMS.

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Exhibit 3 (continued)

Current Team

1. Andreas Bechtolsheim, Ph.D. student at Stanford University, the principal designer of the SUN workstation while at Stanford. Responsibilities: engineering and development, production engineering, system engineering, new product definition.

2. Vinod Khosla, M.S., MBA Stanford University, was a member of the founding team of a successful CAD startup (Daisy Systems). Responsibilities: overall management, finance, strategic planning.

3. Scott McNealy, MBA Stanford University, is currently Director of Operations at Onyx Systems. Responsibilities: Operations.

The team will be expanded over the next few month, concentrating on the areas of marketing and software. In addition, the company can draw on consultants that are currently with VLSI SYSTEMS:

1. Vaughan Pratt, Professor of Computer Science, Stanford University. Vaughan has managed the SUN project at Stanford over the last year and was the main implementor of the SUN software environment.

2. Forest Baskett, Professor of Computer Science, Stanford University, Forest started the SUN workstation project and defined some of its key architectural features. Forest is currently with Xerox Corporation.

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Exhibit 3 (continued)

Appendix A: Costs

Product:

SUN-1: Current SUN workstation.
SUN-2: Single Board Version of SUN-1.
SUN-3: Custom LSI Version of SUN-2.

Direct Material Costs (qty 1000/year):

	SUN-1	SUN-2	SUN-3
Electronics (ICs):	$1100	$1000	$800
Other Parts:			
Monitor:	$400	$350	$300
Keyboard	$120	$100	$70
Mouse	$180	$100	$80
Power Supply	$100	$50	$50
Packaging	$600	$300	$200
	---	---	---
Subtotal	$1400	$800	$700
Total Material Cost:	$2500	$1800	$1500
Direct Labor (10%)	$250	$180	$150
Indirect Labor (10%)	$250	$180	$150
Total Manuf. Cost	$3000	$2240	$1800
Average Selling Price:	$7000	$5000	$4000

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Exhibit 3 (continued)

Appendix B: Financial Requirements February 1982 - May 1982

Salaries (6 people at $ 2,000/month + benefits)	$ 60,000
Marketing, Advertising, and Sales	$ 15,000
Hardware Development	$ 25,000
Software Development (UNIX license fees)	$ 60,000
G & A (including travel and legal)	$ 15,000
Rent & Utilities	$ 5,000

Total Expenses:	$ 180,000
Inventory (20 systems at $3,500)	$ 70,000

Total Cash Requirement:	$ 250,000

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Exhibit 3 (continued)

HW Hardware Eng.
ME Manuf. Eng
SW Software Eng
TW Technical writer
AE Applications Eng.
MB Marketing
GA General & Admin.
MO Manufacs Operations

Appendix C

○ EARLIEST REQUIRED

ENGINEERING DEVELOPMENTS REQUIRED

	START DATE	RESP. INDIVID.	COLLAB. WITH
Hardware (HW)			
Mouse & keyboard			
Foreign version of SUN-1	5/1	HW2	
10 MB Ethernet & Transceiver	5/1	HW1	
Single disk workstation integration	2/15	HW1	SW1
Color graphics	8/1	HW3	HW1/MK2
Product family (cost reduction, options etc.)	7/1	HW1	HW2/MK2
File Server / gateway hardware	8/1	HW1	SW2
Floating point processor	7/1	HW2	HW1
Patent filing	3/1	HW1	
Ongoing debug & maintenance	5/1	HW2	HW1
Manufacturing Engineering			
EMI/RFI & safety requirements	5/1	ME1	Consultant
Acceptance tests (at all levels)	4/1	ME1	HW2
Primary & Second sources "Vendor Selections"	2/15	ME1	HW1
Diagnostics	3/1	HW2	ME1
Documentation	3/1	HW2/ME1	HW1
Parts list & standard costs	3/1	ME1	HW1
Packaging	2/15	ME1	HW1/MK1
Software (SW)			
Single user UNIX	2/15	SW1	SW2/HW1
Distributed UNIX	2/15	SW2	SW1/HW1
Software development aids (PWB)	6/1	SW5	SW1/SW2
Text editor	5/1	SW4	AE1/MS2
Graphics editor	5/1	SW9	AG1/MS2
Benchmarks	6/1	AE1	MS2
File server / gateway software	3/15	SW1	HW1/SW2
Documentation	5/1	TW1	
Programming languages	4/1	SW3	
Software configuration management	10/1	SW6	

Mouse Programming.
Networking Software
Graphics Software
(UNIX (single user/distr.)
Fileserver

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Exhibit 3 (continued)

ORGANIZATIONAL DEVELOPMENTS REQUIRED

General & Administrative (GA)	START DATE	RESP. INDIV.	COLLAB. WITH
Form corporation (select names, lawyers, auditors, bankers, sort licenses)	2/8	GA1/HW1	
Seed financing	2/15	GA1/HW1	
Full financing	4/15	GA1/HW1	
Business licenses (county, State, Federal, sales) & filings (taxes etc).	2/15	GA1	Consultant
Establish payroll, benefit, & insurance programs	3/1	GA--	GA1
Investor relations /reports	2/8	GA1/HW1	
Recruiting	2/8	GA1/HW1	
Accounting system (cost/general).	3/1	GA--	GA1/MO1
Budgets, controls, & planning (cash flows, forecasts, staffing)	2/15	GA--	All others
Facilities	2/15	GA1	MO
Banking (cash mgmt., credit lines, etc.)	2/15	GA1	
Marketing & Sales (MS)			
Sell sample evaluation & development units	3/15	MS1	
Brochures & advertising material	3/15	MS1	Consultant
OEM, research & university sales	5/1	MS--	
Sales training & tools (slide presentation etc)	4/1	MS.	MS2
Competitive analysis (pricing, new products etc).	6/1	MS3	MS1
Public relations & exposure (announcements, articles, conferences)	6/1	MS3	Consultant
Sales program (OEM contracts, licenses, sales compensation plan etc)	4/1	MS1	
Trade shows	4/1	MS3/MS1	
Establish sales offices (national, European, Japan trading Co.)	7/1	MS1	MS--
Order administration (schedul., credit check, etc.)	7/1	MS1	GA1/MO1

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Exhibit 3 (continued)

Manufacturing Operations	START DATE	RESP. INDIV.	COLLAB. WITH
Production line organiz. (work orders, assembly flow, scheduling etc)	3/1	MO1	ME1
Vendor qualification	3/1	MO1	ME1
Quality assurance (test & accep. procedures, ECO's etc.)	5/1	MO1	ME1
Materials systems (part no., inventory control & stockroom orgz., returns etc)	4/1	MO1/MO2	GA1
MRP	4/1	MO1/MO2	
Purchasing procedures & operations	4/1	MO1/MO3	GA1
Shipping / receiving procedures & operations	5/1	MO1/MO4	
Board debug / repair			
System assembly			

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Exhibit 3 (continued)

Appendix D

STAFFING PLAN 1982-83

		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUN
Software	Engineers		2	2	3	4	5	5	5	5	5	5	5	7	7	7	10	10	10
	Support		0	0	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
														New projects					
Hardware	Engineers		1	1	1	2	2	3	3	3	3	3	3	3	3	3	3	3	3
Manuf. Eng.	Engineers		1	1	1	1	1	1	1	1	1	1	1	1	1	1	2	2	2
Mkt. & Sales	Management		0	1	1	1	2	2	2	2	2	2	2	3	3	3	5	5	5
	Sales		0	0	0	0	0	2	2	2	3	3	3	4	4	4	7	7	7
	Applicat. Eng		0	0	0	0	0	1	1	1	2	2	2	2	2	2	3	3	3
	Support		0	0	0	0	1	1	1	1	2	2	2	3	3	3	4	4	4
														Start European oper.					
Gen. & Admin.	Management		1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
	Finan./Acctg		0	0	0	0	0	1	1	1	1	1	1	2	2	2	2	2	2
	Personnel		0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1
	Support		0	1	1	1	1	1	1	1	3	3	3	4	4	4	5	5	5
Manuf. Oper.	Management		0	1	1	1	1	2	2	2	3	3	3	3	3	3	3	3	3
	Buyer/Planner		0	0	0	0	1	1	1	1	2	2	2	2	2	2	3	3	3
	Technicians		0	0	1	1	1	1	1	1	2	2	2	4	4	4	5	5	5
	Hourly		0	0	0	0	0	1	2	2	3	4	4	4	4	5	5	5	5
	Clerical		0	0	0	0	0	1	1	1	1	2	2	2	2	2	3	3	3
	QA Eng							1	1	1	1	1	1	1	1	1	1	1	1
TOTAL EMPLOYEES			5	7	9	11	17	26	27	27	38	40	40	49	50	51	66	66	66
# OF UNITS MODIFIED			—	—	—	—	—	5	15	25	30	35	40	45	50	55	60	65	75

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